Status	Finished
Started	Sunday, 18 October 2026, 9:46 PM
Completed	Sunday, 18 October 2026, 9:52 PM
Duration	6 mins 41 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is very small
b. When the list contains duplicate elements
c. When the list is sorted ✓
d. When the list is unsorted

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It can only be applied to large lists
b. It is slower than linear search for small lists
c. It does not work with negative numbers
d. It requires the list to be sorted ✓

Question **3**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 90 and 99 ✓
b. 89 and 94
c. 89 and 99
d. 90 and 94

Question **4**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

a. Sorting elements in a list
b. Inserting elements into a list
c. Determining whether an element is present in a list ✓
d. Deleting elements from a list

Question **5**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O(1)
b. O🥐 ✓
c. O(n log n)
d. O(log n)

Question **6**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Binary search ✓
b. Sequential search
c. Linear search
d. Hash search

Question **7**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By dividing the list length by two ✓
b. By starting from the first element
c. By using a hash function
d. By comparing each element sequentially

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. The list must be sorted

b. There must be a mechanism to delete and/or insert elements in the list ✓

c. There should be direct access to the middle element in any sublist

d. Number values should only be present

Number values should only be present

Question **9**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search requires the list to be sorted.

b. Linear search divides the list into halves.

c. Linear search is more efficient than binary search on large lists.

d. Linear search can be applied to both sorted and unsorted lists. ✓

Question **10**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

a. Quick search

b. Linear search ✓

c. Merge search

d. Bubble search

Question **11**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the left sublist ✓

b. In the entire list

c. At the beginning of the list

d. In the right sublist

Question **12**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

- a. When the list is small and unsorted ✓
- b. When the list is large and unsorted
- c. When the list is large and sorted
- d. When the list is small and sorted

Question **13**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

- a. When the item is the last element in the array
- b. When the item is somewhere in the middle of the array ✓
- c. When the item is not the array at all
- d. Item is the last element in the array or item is not there at all

Question **14**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

- a. The first element is returned
- b. The last element is returned
- c. An error is raised
- d. The search is considered unsuccessful ✓

Question **15**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

- a. Binary search
- b. Interpolation search
- c. Hash search
- d. Linear search ✓